CANBUS Protocol

of

High Voltage system

Change record				
Date	**Version**	**Chapter**	**Description**	**Author**
2016/7/19	1.00		First Version.	邹慧兴
2016/8/15	1.01		1、Add sleep wake up control/增加休眠唤醒控制/ 2、Add relay faulty check./ 增加继电器检测故障	姜炜
2016/8/17	1.02		Add charge/discharge command./ 增加充放电命令	姜炜
2016/11/15	1.03		Add forced charge and balance charge flag bit /增加强 充、均充标志位	姜炜
2016/12/04	1.04		Add broadcast to get information of battery cell and module /增加广播获取单芯及模块信息	邹慧兴
2016/12/19	1.05		Add voltage, temperature value, alarm and protection of module /增加电池模块电压温度值及告警保护信息 告警保护信息	姜炜
2016/12/22	1.06		Add system equipment information/ 增加系统装备信息 Add commando of relay force to break/ 增加继电器强 制断开命令	姜炜
2017/01/03	1.07		New composing for external PCS communication or internal BMCU communication/ 调整文本顺序，便于 区分对外 PSC 通信或者内部 BMCU 模块通信	邹慧兴
2017/01/12	1.08		Add charge forbidden and discharge forbidden mark. 增加禁止充电，禁止放电标志	邹慧兴

2017/08/22	1.09		Add mark of connect multi-racks(pile) in parallel 增加并柜标志	秦威
2017/9/22	1.10		Add mark of connect multi-rack(pile) in parallel in all broadcast frames 所有广播帧中增加并柜标志	姜炜
2017/11/08	1.11		Add mark of “other error” 增加“其他故障”标志	
2018/03/27	1.12		去掉并柜标志，去除无关指令，此协议只做对外通信对 接使用。 Delete mark of connect multi-racks(pile) in parallel, delete irrelevant internal communication command, this protocol is for customer only.	姜炜
2018/04/02	1.13		增加临时屏蔽”外部通信故障”功能 Add function of keep relay on even there is no communication.	姜炜
2018/06/12	1.14	1.2 Page8	系统装备信息响应数据 Revise System Device information’s response data 1. 调整系统装备信息中的“硬件版本信息”和“软件版 本 信 息 ” Revise the “hardware version info.” and “software version info.” In the system device information data 2. 系统装备信息中电池容量信息由一个字节改为 2 字 节表示，原来只支持最大 255AH，改后最大 65535AHRevise the system device information’s date from 1bit to 2bits, so that the battery capacity can be showed from 255Ah to 65535Ah	邹慧兴
2018/09/25	1.15	Table 2	增加电池损坏故障位 Add Battery cell error mark	
2018/11/06	1.16	Table 5	增加故障扩展位： 1. 增加关机电路等故障位 Add Shutdown circuit error mark 2. 增加 BIMC 故障位 Add BMIC error mark 3. 增加内部总线异常故障位 Add Internal bus error mark 4. 增加开机自检异常故障位 Add Self-test error	姜炜

2020/09/09	1.17	Page9	增加电池制造厂商信息 ASCII 码： Add ASCII code for battery manufacturer information: 定义 CANID:0x42F0 Define CANID: 0x42F0 Byte0~Byte7: For exzample: Dyness Byte0=’ D’ ;Byte1=’ y’ ;Byte2=’ n’ ;Byte3=’ e’ ;Byte4=’ s’ ;Byte5=’ s’ ;	Lane
2020/09/09	1.17	Page10	增加电池序列号信息 ASCII 码： Add ASCII 码 code for battery serial number information: 定义 CANID:0x42E0 Define CANID: 0x42E0 Byte0~Byte7: For exzample: 12345678 Byte0=’ 1’ ;Byte1=’ 2’ ;Byte2=’ 3’ ;Byte3=’ 4’ ;Byte4=’ 5’ ;Byte5=’ 6’ ， Byte5=’ 7’ Byte5=’ 8’ ;	Lane
2020/09/16	1.18	Page10	增加电池型号信息： Add Battery Type information: 定义 CANID:0x4300 Define CANID: 0x4300 Byte0 ： Battery Type	Lane
2021/03/13	1.19	Page13	增加设置系统和模组信息功能 Added setting system and module information function 定义 CANID：0x8220 Define CANID: 0x8220 Define Byte0：system model； Define Byte1：total module numbers； Define Byte2：module capacity；	Lane
2022/02/14	1.20		Added the support for standard frames, When the background monitoring software sends with the standard frame ID, the battery system replies with the standard frame ID 增加对标准帧的支持,上位机使用标准帧 ID 发送时，电池系统则使用标准帧 ID 回复；	Lane

CAN 总线规格： CANBUS frame format:

采用 29 位标识符的扩展帧格式传输，总线传输速率为 500kbps.

The 29 bits identifier is used to transfer the extended frame format, and the bus transmission rate is 500kbps.

上位机设备发送 29 位 ID 的扩展帧，电池回复 29 位 ID 的扩展帧数据，上位机发送 11 位 ID 的标准帧，则电池也回复 11 位 ID 的标准帧数据。

The host device sends the 29-bit ID extended frame，the battery respond the 29-bit extended frame data，The host The host device sends the 11-bit ID standard frame，the battery also respond the 11-bit standard frame data.

通信方式： Communication Mode:

上位机设备发送查询或控制指令后，电池组设备响应数据。

After the host device sends the check or control command, the battery system responds data.

Report Sending is LSB. 发送为低字节在前

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1 查询指令 Query Instruction

1.1 总体信息 Ensemble information

- 上位机设备发送数据：Host device sending data:

CAN ID：0x4200(29 bits identifier) or 0x420(11 bits identifier) （为广播帧）(this is Broadcast Frame)

Byte0	0	Ensemble Information 总体信息
Byte1	Reserve	
Byte2	Reserve	
Byte3	Reserve	
Byte4	Reserve	
Byte5	Reserve	
Byte6	Reserve	
Byte7	Reserve	

- 电池组设备响应数据：The battery pile device responds the following data:

CAN ID:0x4210+Addr (29 bits identifier)（设备地址 Addr= 0~F，最多 16 台设备并联通信）

(Equipment Address: Addr. = 0~F, MAX 16 equipments can be parallel communicated.)

CAN ID:0x421 (11 bits identifier)

Byte0	电池组总电压 Battery Pile Total Voltage	Resolution: 0.1V
Byte1		Offset: 0
Byte2	电池组电流 Battery Pile Current	Resolution: 0.1A
Byte3		Offset: -3000A
Byte4	主控温度 second level BMS Temperature	Resolution: 0.1 °C
Byte5		Offset: -100 °C
Byte6	SOC	Resolution: 1%, Offset: 0
Byte7	SOH	Resolution: 1%, Offset: 0

CAN ID:0x4220+Addr (29 bits identifier)（设备地址 Addr=0~F，最多 16 台设备并联通信）
(Equipment Address: Addr=0~F, MAX 16 equipments can be parallel communicated.)
CAN ID:0x422 (11 bits identifier)

Byte0	充电截止电压 Charge Cutoff Voltage	Resolution: 0.1V
Byte1		Offset: 0
Byte2	放电截止电压 Discharge Cutoff Voltage	Resolution: 0.1V
Byte3		Offset: 0
Byte4	最大充电电流 MAX Charge Current	Resolution: 0.1A
Byte5		Offset: -3000A
Byte6	最大放电电流 MAX Discharge Current	Resolution: 0.1A
Byte7		Offset: -3000A

CAN ID:0x4230+Addr (29 bits identifier)（设备地址 Addr=0~F，最多 16 台设备并联通信）
(Equipment Address: Addr=0~F,MAX 16 equipments can be parallel communicated.)
CAN ID:0x423 (11 bits identifier)

Byte0	最高单体电池电压	Resolution: 0.001V
Byte1	MAX Single Battery Cell Voltage	Offset: 0
Byte2	最低单体电池电压	Resolution: 0.001V
Byte3	MIN Single Battery Cell Voltage	Offset: 0
Byte4	最高单体电池电压编号	Resolution: 1
Byte5	MAX Single Battery Cell Voltage Number	Offset: 0
Byte6	最低单体电池电压编号	Resolution: 1
Byte7	MIN Single Battery Cell Voltage Number	Offset: 0

CAN ID:0x4240+Addr (29 bits identifier)（设备地址 Addr=0~F，最多 16 台设备并联通信）
(Equipment Address: Addr=0~F,MAX 16 equipments can be parallel communicated.)
CAN ID:0x424(11 bits identifier)

Byte0	最高单体电池温度	Resolution: 0.1°C
Byte1	MAX Single Battery Cell Temperature	Offset: -100
Byte2	最低单体电池温度	Resolution: 0.1 °C
Byte3	MIN Single Battery Cell Temperature	Offset: -100
Byte4	最高单体电池温度编号 MAX Single	Resolution: 1
Byte5	Battery Cell Temperature Number	Offset: 0
Byte6	最低单体电池温度编号 MIN Single Battery	Resolution: 1
Byte7	Cell Temperature Number	Offset: 0

CAN ID:0x4250+Addr (29 bits identifier)（设备地址 Addr=0~F，最多 16 台设备并联通信）
(Equipment Address: Addr=0~F,MAX 16 equipments can be parallel communicated.)
CAN ID:0x425 (11 bits identifier)

Byte0	基本状态 Basic Status	详见附表一 See Table 1 for details.
Byte1	循环周期 Cycle Period	
Byte2		
Byte3	故障 Error	详见附表二 See Table 2 for details.
Byte4	告警 Alarm	详见附表三 See Table 3 for details.
Byte5		
Byte6	保护 Protection	详见附表四 See Table 4 for details.
Byte7		

Table 1: Basic Status

Bit7	Reverse
Bit6	Reverse
Bit5	Reverse
Bit4	0 ：Null;　1：请求均充/Balance charge request
Bit3	0 ：Null;　1：请求强充/Forced charge request
Bit2	0: Sleep, 1: Charge, 2: Discharge, 3: Idle, 4~7: Reserve 0：休眠， 1 ：充电，2：放电， 3 ：搁置， 4~7：保留
Bit1	
Bit0	

Table 2: Fault

Bit7	Other error 其他故障(具体见故障扩展)
Bit6	电池损坏故障(电池过放等原因导致) / Battery cell error
Bit5	RELAY_ERR/ 继电器检测故障 Relay Check Error
Bit4	RV_ERR/ 输入反接故障 Input transposition Error
Bit3	DCOV_ERR/ 输入过压故障 Input Over Voltage Error
Bit2	IN_COMM_ERR/ 内部通信故障 Internal Communication Error
Bit1	TMPR_ERR/ 温度传感器故障 Temperature Sensor Error
Bit0	VOLT_ERR/ 电压传感器故障 Voltage Sensor Error

Table 3: Alarm

Bit15	Reserve
Bit14	Reserve
Bit13	Reserve
Bit12	Reserve
Bit11	MHV: 电池模块高压告警 Module High Voltage Alarm
Bit10	MLV: 电池模块低压告警 Module Low Voltage Alarm
Bit9	DOCA: 电池组放电过流告警 Discharge Over Current Alarm
Bit8	COCA: 电池组充电过流告警 Charge Over Current Alarm
Bit7	DHT: 放电高温告警 Discharge Cell High Temperature Alarm
Bit6	DLT: 放电低温告警 Discharge Cell Low Temperature Alarm
Bit5	CHT: 充电高温告警 Charge Cell High Temperature Alarm
Bit4	CLT: 充电低温告警 Charge Cell Low Temperature Alarm
Bit3	PHV: 电池组充电高压告警 Charge system High Voltage Alarm
Bit2	PLV: 电池组放电低压告警 Discharge system Low Voltage Alarm
Bit1	BHV: 电池单体高压告警 Single Cell High Voltage Alarm
Bit0	BLV: 电池单体低压告警 Single Cell Low Voltage Alarm

Table 4: Protection

Bit15	Reserve
Bit14	Reverse
Bit13	Reverse
Bit12	BUV2: 电池单体二级欠压保护 Module 2nd level under voltage protection
Bit11	MOV: 电池模块过压保护 Module Over Voltage Protect
Bit10	MUV: 电池模块欠压保护 Module Under Voltage Protect
Bit9	DOC: 电池组放电过流保护 Discharge Over Current Protect
Bit8	COC: 电池组充电过流保护 Charge Over Current Protect
Bit7	DOT: 放电高温保护 Discharge Cell Over Temperature Protect
Bit6	DUT: 放电低温保护 Discharge Cell Under Temperature Protect
Bit5	COT: 充电高温保护 Charge Cell Over Temperature Protect
Bit4	CUT: 充电低温保护 Charge Cell Under Temperature Protect
Bit3	POV: 电池组充电高压保护 Charge system Over Voltage Protect
Bit2	PUV: 电池组放电低压保护 Discharge system Under Voltage Protect
Bit1	BOV: 电池单体高压保护 Single Cell Over Voltage Protect
Bit0	BUV: 电池单体低压保护 Single Cell Under Voltage Protect

CAN ID:0x4260+Addr (29 bits identifier)（设备地址 Addr=0~F，最多 16 台设备并联通信）
(Equipment Address: Addr=0~F,MAX 16 equipments can be parallel communicated.)

CAN ID:0x426 (11 bits identifier)

Byte0 Byte1	Module Max. Voltage 最高电池模块电压	Resolution: 0.001V Offset: 0
Byte2 Byte3	Module Min. Voltage 最低电池模块电压	Resolution: 0.001V Offset: 0
Byte4 Byte5	Module Max. Voltage Number 最高电池模块电压编号	Resolution: 1 Offset: 0
Byte6 Byte7	Module Min. Voltage Number 最低电池模块电压编号	Resolution: 1 Offset: 0

CAN ID:0x4270+Addr(29 bits identifier)（设备地址 Addr=0~F，最多 16 台设备并联通信）
(Equipment Address: Addr=0~F,MAX 16 equipments can be parallel communicated.)

CAN ID:0x427 (11 bits identifier)

Byte0 Byte1	Module Max. Temperature 最高电池模块 温度	Resolution: 0.1°C Offset: -100
Byte2 Byte3	Module Min. Temperature 最低电池模块温 度	Resolution: 0.1°Ce Offset: -100
Byte4 Byte5	Module Max. Temperature Number 最高电池模块温度编号	Resolution: 1 Offset: 0
Byte6 Byte7	Module Min. Temperature Number 最低电池模块温度编号	Resolution: 1 Offset: 0

CAN ID:0x4280+Addr(29 bits identifier)（设备地址 Addr=1~F，最多 15 台设备并联通信）
(Equipment Address: Addr=1~F, MAX 15 equipments can be parallel communicated.)

CAN ID:0x428 (11 bits identifier)

Byte0	Charge forbidden mark 禁止充电标志	0xAA 有效，其它值无效 0xAA：effect; other value: NULL
Byte1	Discharge forbidden mark 禁止放电标志	0xAA 有效，其它值无效 0xAA：effect；other value: NULL
Byte2	Reserve	
Byte3	Reserve	
Byte4	Reserve	
Byte5	Reserve	
Byte6	Reserve	
Byte7	Reserve	

CAN ID:0x4290+Addr(29 bits identifier)（设备地址 Addr= 0~F，最多 16 台设备并联通信）

(Equipment Address: Addr. = 0~F, MAX 16 equipments can be parallel communicated.)

CAN ID:0x429 (11 bits identifier)

Byte0	故障扩展 1	详见附表五 See Table 5 for details.
Byte1	Reserve	
Byte2	Reserve	
Byte3	Reserve	
Byte4	Reserve	
Byte5	Reserve	
Byte6	Reserve	
Byte7	Reserve	

Table 5: 故障扩展 1

Bit7	Reserve
Bit6	Reserve
Bit5	Reserve
Bit4	Reserve
Bit3	开机自检异常/ Self-test error
Bit2	内部总线异常/ Internal bus error
Bit1	BMIC 异常/BMIC error
Bit0	关机电路异常/ Shutdown circuit error

CAN ID:0x42E0+Addr(29 bits identifier)（设备地址 Addr= 0~F，最多 16 台设备并联通信）

(Equipment Address: Addr. = 0~F, MAX 16 equipments can be parallel communicated.)

CAN ID:0x42E (11 bits identifier)

Byte0	Series-Number-ASCII	
Byte1		
Byte2		
Byte3		
Byte4		
Byte5		
Byte6		
Byte7		

CAN ID:0x42F0+Addr (29 bits identifier)（设备地址 Addr= 0~F，最多 16 台设备并联通信）(Equipment Address: Addr. = 0~F, MAX 16 equipments can be parallel communicated.)

CAN ID:0x42F (11 bits identifier)

Byte0	Manufacturer-Name-ASCII	
Byte1		
Byte2		
Byte3		
Byte4		
Byte5		
Byte6		
Byte7		

CAN ID:0x4300+Addr (29 bits identifier)（设备地址 Addr= 0~F，最多 16 台设备并联通信）(Equipment Address: Addr. = 0~F, MAX 16 equipments can be parallel communicated.)

CAN ID:0x430 (11 bits identifier)

Byte0	System Type	
Byte1	Reversed	
Byte2	Reversed	
Byte3	Reversed	
Byte4	Reversed	
Byte5	Reversed	
Byte6	Reversed	
Byte7	Reversed	

System Type

Battery Name	Define
Hpowercube 10.0/GTI-10	0x01
Hpowercube 16.0/GTI-16	0x02
Tower T7	0x03
Tower T10	0x04
Tower T14	0x05
Tower T17	0x06
Tower T21	0x07

1.2 系统装备信息 system equipment information

0x4200 (29 bits identifier) or 0x420 (11 bits identifier) (is broadcast frame)（为广播帧）

Byte0	2	2 is system equipment information 系统装备信息
Byte1	Reserve	
Byte2	Reserve	
Byte3	Reserve	
Byte4	Reserve	
Byte5	Reserve	
Byte6	Reserve	
Byte7	Reserve	

- Battery System Response Data: 电池组设备响应数据：

CAN ID: 0x7310+Addr(29 bits identifier) (设备地址 Addr= 0~F，最多 16 台设备并联通信) (Equipment Address: Addr. = 0~F, MAX 16 equipments can be parallel communicated.)

CAN ID: 0x731 (11 bits identifier)

Hardware Version: such as V2.1
Software Version: such as V1.2

Byte0	Hardware Version 硬件版本	0: Null; 1: ver. A; 2: ver. B; Others: Reserve. 0:无效; 1:A 版本; 2:B 版本 ; 其他:预留
Byte1	reserve	
Byte2	Hardware Version-V 硬件版本-V	0x02
Byte3	Hardware Version-R 硬件版本-R	0x01
Byte4	Software Version-V 软件版本-V （主版本 Major）	0x01
Byte5	Software Version-V 软件版本-V （子版本 Minor）	0x02
Byte6	Software Version-开发主版本	
Byte7	Software Version 开发子版本	

CAN ID: 0x7320+Addr(29 bits identifier)（设备地址 Addr= 0~F，最多 16 台设备并联通信）
(Equipment Address: Addr. = 0~F, MAX 16 equipments can be parallel communicated.)
CAN ID: 0x732 (11 bits identifier)

Byte0 Byte1	Battery Module Qty. 电池总数	
Byte2	Battery Module in series Qty. 串联电池模块个数	
Byte3	Cell Qty. in battery module 模块中电池个数	
Byte4 Byte5	Voltage Level 电压平台	Resolution: 1V Offset: 0
Byte6 Byte7	AH number AH 数	Resolution: 1AH Offset: 0

2 控制指令 Control Command

2.1 休眠唤醒控制 Sleep / Awake Command

Addr=0~F, MAX 16 equipments can be parallel communicated.)
不支持广播 Broadcast not supported
上位机设备发送数据： Host device send data:

CAN ID: 0x8200+Addr(29 bits identifier)（设备地址 Addr= 0~F，最多 16 台设备并联通信）
(Equipment Address: Addr. = 0~F, MAX 16 equipments can be parallel communicated.)
CAN ID: 0x620 (11 bits identifier)

Byte0	休眠唤醒控制 Sleep Awake Control	0x55: 控制设备进入休眠状态 Control device enter sleep status; 0xAA: 控制设备退出休眠状态 Control device quit sleep status; Others: Null 其他：无效
Byte1	Reserve	0
Byte2	Reserve	0
Byte3	Reserve	0
Byte4	Reserve	0
Byte5	Reserve	0
Byte6	Reserve	0
Byte7	Reserve	0

电池组无回复 No response from batter

2.2 充电放电命令 Charge/Discharge Command

不支持广播 Broadcast not supported

上位机设备发送数据： Host device send data:

CAN ID: 0x8210+Addr(29 bits identifier)（设备地址 Addr= 0~F，最多 16 台设备并联通信）(Equipment Address: Addr. = 0~F, MAX 16 equipments can be parallel communicated.)

CAN ID: 0x821 (11 bits identifier)

Byte0	充电命令 Charge Command	0xAA: effect; Others: Null (* Note 1)
Byte1	放电命令 Discharge Command	0xAA: effect; Others: Null (* Note 2)
Byte2	Reserve	
Byte3	Reserve	
Byte4	Reserve	
Byte5	Reserve	
Byte6	Reserve	
Byte7	Reserve	

电池组无回复 No response from battery.

*Note:

1. 充电命令：当电池处于欠压保护状态时，继电器断开，当 EMS 或 PCS 确定要对电池进行充电 时可以发送此命令，电池会闭合主继电器。若电池已休眠，则需先唤醒。
   Charge Command: When the battery is in under-voltage protection, the relay is open. When EMS or PCS is going to charge the battery, send this command, then the battery will close the main relay. If the battery is in sleep status, wake up first then use this command.
2. 放电命令：当电池处于过压保护状态时，继电器断开，此时 EMS 或 PCS 确定要对电池进行放 电时， 可发送此命令， 电池会闭合主继电器。若电池已休眠， 则需先唤醒。
   Discharge Command: When the battery is in over-voltage protection, the relay is open. When EMS or PCS is going to discharge the battery, send this command, then the battery will close the main relay. If the battery is in sleep status, wake up first then use this command.

2.3 设置系统型号及模组信息 Set system model and module information

不支持广播 Broadcast not supported

上位机设备发送数据： Host device send data:

CAN ID: 0x8220+Addr(29 bits identifier) (设备地址 Addr= 0~F，最多 16 台设备并联通信)

(Equipment Address: Addr. = 0~F, MAX 16 equipments can be parallel communicated.)

CAN ID: 0x622 (11 bits identifier)

Byte0	系统型号 system model	See table below
Byte1	模组总数量 total number of modules	0x02~0x06
Byte2	模组容量 module capacity	Resolution: 1AH Offset: 0 default 37Ah（0x25）
Byte3	Reserve	
Byte4	Reserve	
Byte5	Reserve	
Byte6	Reserve	
Byte7	Reserve	

电池组无回复 No response from battery.

System model	Define
Tower T7	0x07
Tower T10	0x0A
Tower T14	0x0E
Tower T17	0x11
Tower T21	0x15

2.4 临时屏蔽“通信故障”指令 Temporary masking “external communication error” command

CAN ID: 0x8240+Addr(29 bits identifier)（设备地址 Addr= 0~F，最多 16 台设备并联通信）
(Equipment Address: Addr. = 0~F, MAX 16 equipments can be parallel communicated.)
CAN ID: 0x624 (11 bits identifier)

Byte0	BMS masking “external communication error” 下发“屏蔽外部通信故障”指令	0xAA: effect; Others: Null
Byte1	Reserve	0
Byte2	Reserve	0
Byte3	Reserve	0
Byte4	Reserve	0
Byte5	Reserve	0
Byte6	Reserve	0
Byte7	Reserve	0

电池组回复 Response from battery:
CAN ID: 0x8250+Addr(29 bits identifier)（设备地址 Addr= 0~F，最多 16 台设备并联通信）
(Equipment Address: Addr. = 0~F, MAX 16 equipments can be parallel communicated.)
CAN ID: 0x625 (11 bits identifier)

Byte0	System condition able to act this command or not 系统状态是否符合执行此命令条件	0xAA: 符合，立即执行；OK, will act this commend immediately Others：不执行此命令; won`t act this command
Byte1	Reserve	0
Byte2	Reserve	0
Byte3	Reserve	0
Byte4	Reserve	0
Byte5	Reserve	0
Byte6	Reserve	0
Byte7	Reserve	0

Danger: High Safety Risk from improper use
警示：不正当使用会造成系统严重安全风险。

Note：

After receive this command, BMS will estimate the condition and give reply.

If meet the condition, in 5 minutes, BMS will ignore the “external communication fail” alarm, which means relay will keep ON while no communication between BMS and EMS/PCS.

In this 5 minutes, if there is a protection alarm, BMS will cut off the relay as normal.

当外部设备发起请求时，如电池系统允许执行此动作，则返回正常报文，并且系统将屏蔽 5 分钟“外部通讯故障”功能。在 5 分钟内，继电器将保持闭合。但，当 5 分钟内发生保护时，系统将正常执行保护功能。